Win-06-01b thermal-history modeling (preview)

0.1 Thermal-History Modeling of Win-06-01b

Win-06-01b was collected as float in the Flinders Ranges and has a (U–Th)/He age of 130–180 Ma (Waltenberg, 2012). Apatite fission-track studies across the region record regional cooling from roughly 50–70 °C after the mid-Cretaceous (Gibson and Stüwe, 2000; Mitchell et al., 1998), and Triassic intramontane basins less than 1.5 km thick preserve near-surface Early Mesozoic sections (Mitchell et al., 1998). We therefore ask which burial-and-exhumation histories reproduce both the bulk-normalized ^{4}He/^{3}He spectrum of the water-saturated experiment (Fig. **??**) and the He age, following the ratio-evolution approach of Shuster and Farley (2004).

Diffusion domains. Helium resides in two spherical domains of the same goethite, with a common activation energy and frequency factor: a high-retentivity domain (HRD) of radius a and a low-retentivity domain (LRD) of radius ra ($r < 1$) that holds a fraction f of the uniformly distributed, proton-induced ^{3}He. The HRD follows the Stage 1 Arrhenius regression of the water-saturated experiment (Fig. **??**; $E_a = 157.1$ kJ mol^{-1}, $\ln(D_0/a^2) = 16.16$, D_0/a^2 in s^{-1}), so that

$$\frac{D}{a^2}(T) = \frac{D_0}{a^2}\exp\!\left(-\frac{E_a}{RT}\right), \qquad \left(\frac{D}{a^2}\right)_{\mathrm{LRD}} = r^{-2}\,\frac{D}{a^2}. \tag{1}$$

Diffusion in each domain is described by the non-dimensional time

$$\tau(t) = \int \frac{D(T(t'))}{a^2}\,\mathrm{d}t', \tag{2}$$

and an initially uniform sphere releases the cumulative fraction (Crank, 1979)

$$F(\tau) = 1 - \frac{6}{\pi^2}\sum_{k=1}^{\infty}\frac{1}{k^2}\exp\!\left(-k^2\pi^2\tau\right). \tag{3}$$

Constraining the LRD with ^{3}He. With the HRD's laboratory time τ_i after step i accumulated from Eq. 1 and the heating schedule, the model cumulative ^{3}He release is

$$\Sigma F_{3,i}^{\mathrm{model}} = f\,F\!\left(\tau_i/r^2\right) + (1-f)\,F(\tau_i). \tag{4}$$

We fit f and r by minimizing $\sum_i \left[(\Sigma F_{3,i}^{\mathrm{model}} - \Sigma F_{3,i}^{\mathrm{meas}})/\Sigma F_{3,i}^{\mathrm{meas}}\right]^2$ over all 34 steps, with $f \le 0.075$ as required by the delta plot (Fig. **??**). These two parameters are then held fixed, so the ^{4}He/^{3}He spectrum constrains only the thermal history.

Radiogenic ^{4}He in nature. In each domain, ^{4}He is produced uniformly at rate P and diffuses with a zero-concentration boundary,

$$\frac{\partial C}{\partial t} = \frac{D}{a^2}\,\frac{1}{x^2}\,\frac{\partial}{\partial x}\!\left(x^2\frac{\partial C}{\partial x}\right) + P, \qquad C(1,t) = 0, \quad C(x, t_{\mathrm{form}}) = 0, \tag{5}$$

where x is radial position normalized to the domain radius and t_{form} is the crystallization age. Expanding the concentration in eigenfunctions (Carslaw and Jaeger, 1959), $C(x,t) = \sum_k A_k(t)\,\sin(k\pi x)/x$, each mode obeys

$$\frac{\mathrm{d}A_k}{\mathrm{d}t} = -k^2\pi^2\frac{D}{a^2}A_k + c_k P, \qquad c_k = \frac{2(-1)^{k+1}}{k\pi}, \tag{6}$$

where c_k are the coefficients of a uniform profile. Discretizing the thermal history into intervals Δt (2 Myr) of constant temperature gives the exact update

$$A_k \leftarrow A_k\, e^{-\lambda_k} + c_k P\, \Delta t\, \frac{1 - e^{-\lambda_k}}{\lambda_k}, \qquad \lambda_k = k^2\pi^2\, \frac{D}{a^2}\, \Delta t, \tag{7}$$

which we evaluate with 8000 modes. The ^{4}He content per unit volume is $N = 3\sum_k A_k\,(-1)^{k+1}/(k\pi)$, and, taking U and Th to be distributed like the ^{3}He, the predicted bulk He age is

$$t_{\mathrm{He}} = \frac{(1-f)\,N_{\mathrm{HRD}} + f\,N_{\mathrm{LRD}}}{P}, \tag{8}$$

neglecting parent decay (a $\sim$3% effect over 200 Myr).

Laboratory degassing of the natural profile. In the laboratory each mode decays as $\exp(-k^2\pi^2\tau)$. The HRD's laboratory times τ_i are solved from the measured cumulative ^{3}He through Eq. 4 (the LRD's are τ_i/r^2), which removes any dependence on the heating schedule or on step-to-step kinetic scatter. The predicted spectrum is

$$\frac{R_{\mathrm{step},i}}{R_{\mathrm{bulk}}} = \frac{\Delta N_{4,i}/N_{4,\mathrm{tot}}}{\Delta F_{3,i}}, \tag{9}$$

where $\Delta N_{4,i}$ is the ^{4}He released from both domains in step i and $\Delta F_{3,i}$ is the measured fraction of total ^{3}He released in that step.

Thermal histories and fitting. The thermal history is piecewise linear through (t_{form}, T_s), $(t_{\mathrm{peak}}, T_{\mathrm{max}})$, (50 Ma, T_s), and $(0, T_s)$, with $T_s = 20\,^{\circ}$C: the goethite crystallizes at the surface, is heated during burial to a peak temperature, and is exhumed by the Paleogene, when Paleogene sediments in Flinders Ranges valleys show that basement was exposed (Mitchell et al., 1998). We searched t_{form} from 180 to 210 Ma (5 Myr steps), t_{peak} from 55 Ma to t_{form} (2.5 Myr steps), and T_{max} from 20 to 90 °C (1 °C steps). Each history is scored against the 24 steps with lag-correction bounds and > 0.1 pccSTP ^{3}He by

$$\chi^2 = \sum_i \left(\frac{R_i^{\mathrm{model}} - R_i^{\mathrm{M}}}{\sigma_i} \right)^2, \qquad \sigma_i = \max\left[\tfrac{1}{2}\left(R_i^{\mathrm{UB}} - R_i^{\mathrm{LB}} \right),\ 0.02 \right], \tag{10}$$

where M, LB and UB are the central, lower-bound and upper-bound bulk-normalized ratios of the bounding method. Histories within $\Delta\chi^2 \le 3.53\,\max(1, \chi^2_\nu)$ of the minimum form the joint 1σ region for three parameters. For comparison we also fit isothermal holding (one parameter, τ_{nat}) and a symmetric history heating linearly from 200 to 100 Ma and cooling linearly to the present (one parameter, T_{max}). The kinetic extrapolation uncertainty of the Stage 1 line ($\pm\sim$4 °C near 60 °C) is not propagated.

0.2 Thermal History of Win-06-01b

The two-domain model reproduces the measured cumulative ^{3}He release to 7% (root-mean-square relative misfit), compared with 44% for the Stage 1 line alone. It requires an LRD holding 1.7% of the ^{3}He with $r = 0.0024$, which is degassed within the first $\sim$7 steps and holds negligible radiogenic ^{4}He in nature (Figs. 1 and 2).

With the LRD fixed, burial histories peaking at 60–64 °C between 87.5 and 57.5 Ma fit the spectrum ($\chi^2_\nu \approx 2.2$–2.3) and predict He ages within the measured range for crystallization at 180–200 Ma (Figs. 3–5). The spectrum alone barely distinguishes crystallization ages ($\chi^2_\nu = 2.18$–2.42 across 180–210 Ma); the He age excludes crystallization before ∼200 Ma. The best-fitting age-consistent history crystallizes at 200 Ma, reaches 62 °C at 77.5 Ma and predicts a He age of 179 Ma. These peak temperatures and timing agree with the 50–70 °C cooling after ∼85 Ma inferred from apatite fission tracks (Gibson and Stüwe, 2000) and sit at the <60 °C limit reported near Copley (Mitchell et al., 1998). The symmetric history fits almost as well ($T_{\max} = 60$ °C, $\chi^2_\nu = 2.5$, $t_{\mathrm{He}} = 180$ Ma), whereas isothermal holding is rejected ($\chi^2_\nu = 16$; Fig. 4).

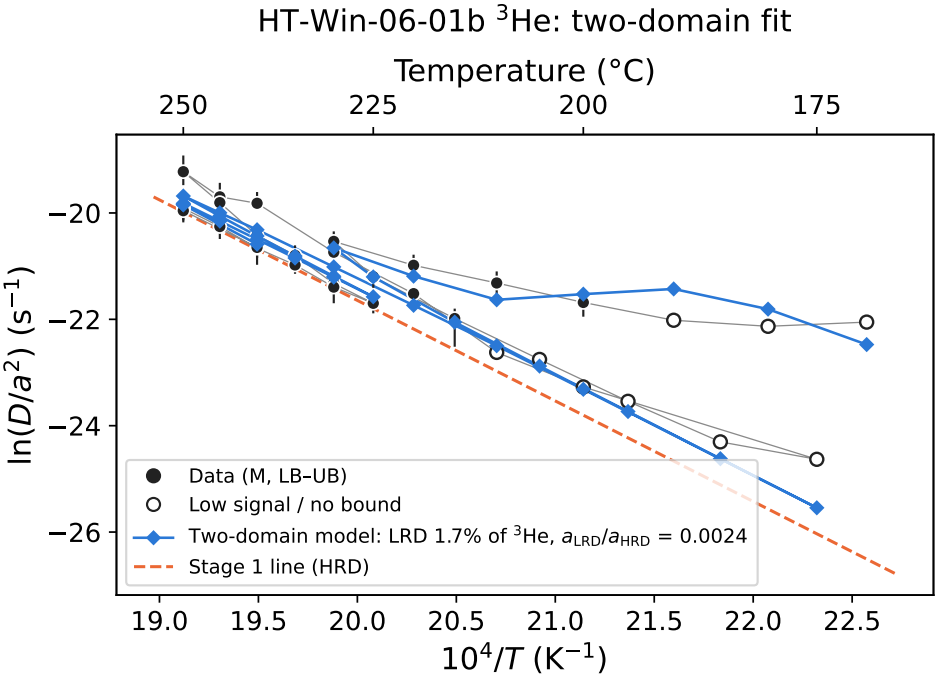


Figure 1: Arrhenius plot of ^{3}He diffusivities in HT-Win-06-01b with the two-domain model. Black circles are bounding-method central values with lower and upper bounds; hollow circles are low-signal steps or steps without lag bounds. Blue diamonds are apparent diffusivities obtained by applying the same single-sphere inversion (Fechtig and Kalbitzer, 1966) to the model's cumulative release (Eq. 4); the dashed line is the Stage 1 regression, which describes the HRD. An LRD holding 1.7% of the ^{3}He with $a_{\mathrm{LRD}}/a_{\mathrm{HRD}} = 0.0024$ reproduces the elevated early diffusivities and the retrograde steps.

References

H. S. Carslaw and J. C. Jaeger. *Conduction of Heat in Solids*. Oxford University Press, Oxford, 2nd edition, 1959.

J. Crank. *The Mathematics of Diffusion*. Clarendon Press, 1979.

H. Fechtig and S. Kalbitzer. The diffusion of argon in potassium-bearing solids. In O. A. Schaeffer

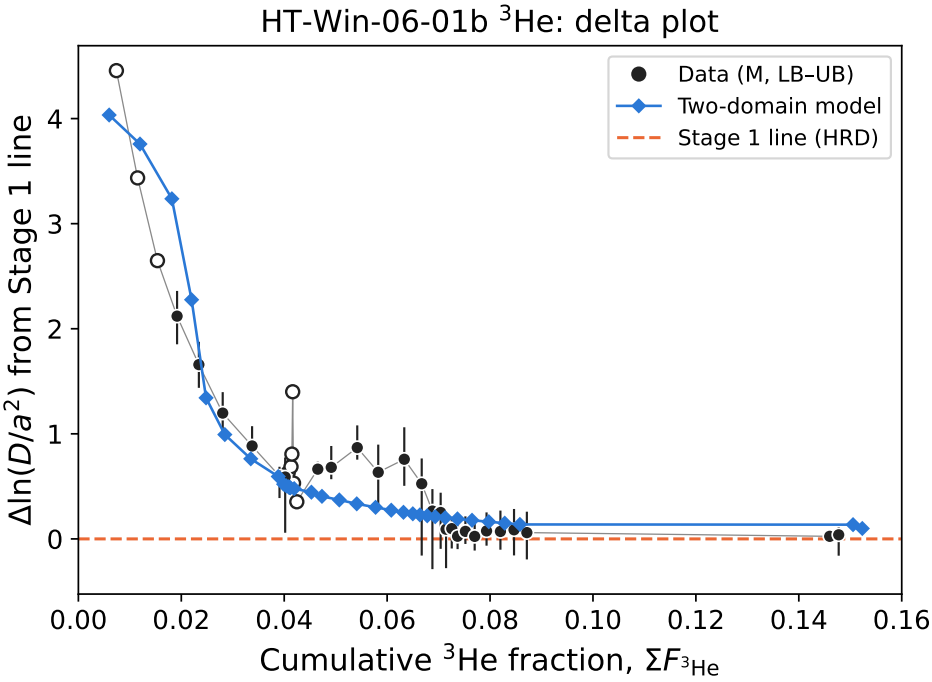


Figure 2: Delta plot of the same data and model: departure of apparent $\ln(D/a^2)$ from the Stage 1 line against cumulative ^{3}He fraction. The model approaches the Stage 1 line as the LRD is exhausted, consistent with an LRD holding no more than 7.5% of the ^{3}He.

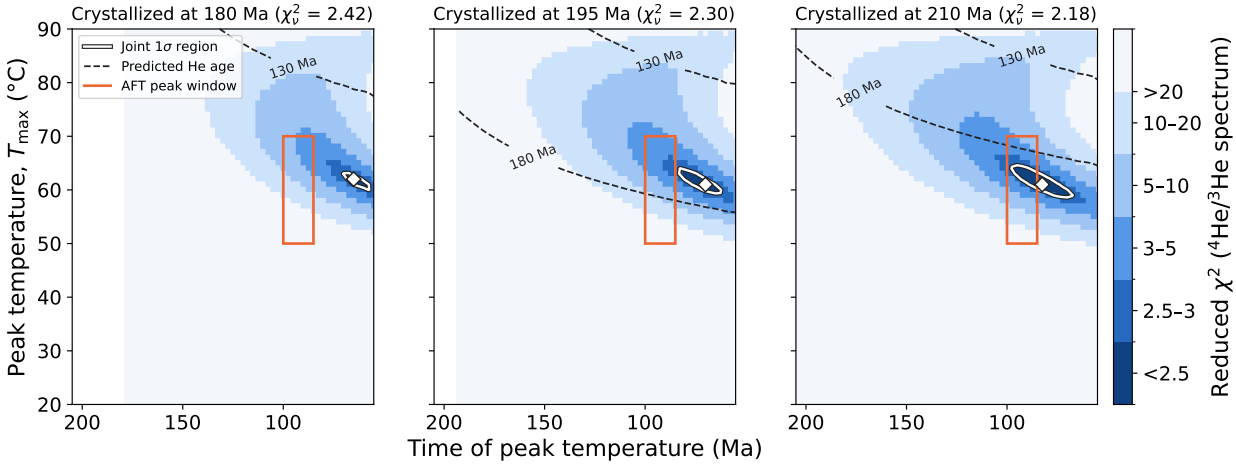


Figure 3: Fit of burial-and-exhumation histories to the ^{4}He/^{3}He spectrum for crystallization at 180, 195 and 210 Ma. Colors show reduced χ^2 as a function of the time and magnitude of peak temperature; white outlines enclose the joint 1σ region across all three parameters, and diamonds mark each panel's best fit. Dashed lines are isopleths of the predicted bulk He age at the limits of the measured range (130 and 180 Ma). Orange boxes show the peak window implied by apatite fission-track data (Gibson and Stüwe, 2000; Mitchell et al., 1998).

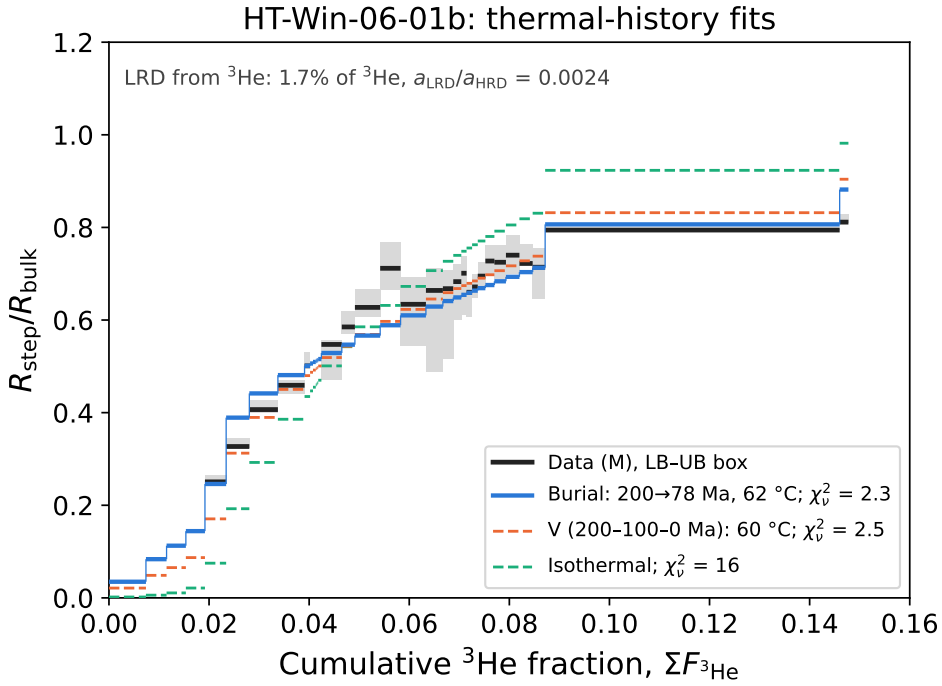


Figure 4: Bulk-normalized ^{4}He/^{3}He spectrum of HT-Win-06-01b (black, with bounding-method LB–UB boxes) compared with the best-fitting age-consistent burial history (blue), the best symmetric 200–100–0 Ma history (orange) and the best isothermal history (green), all computed with the LRD fixed from the ^{3}He fit.

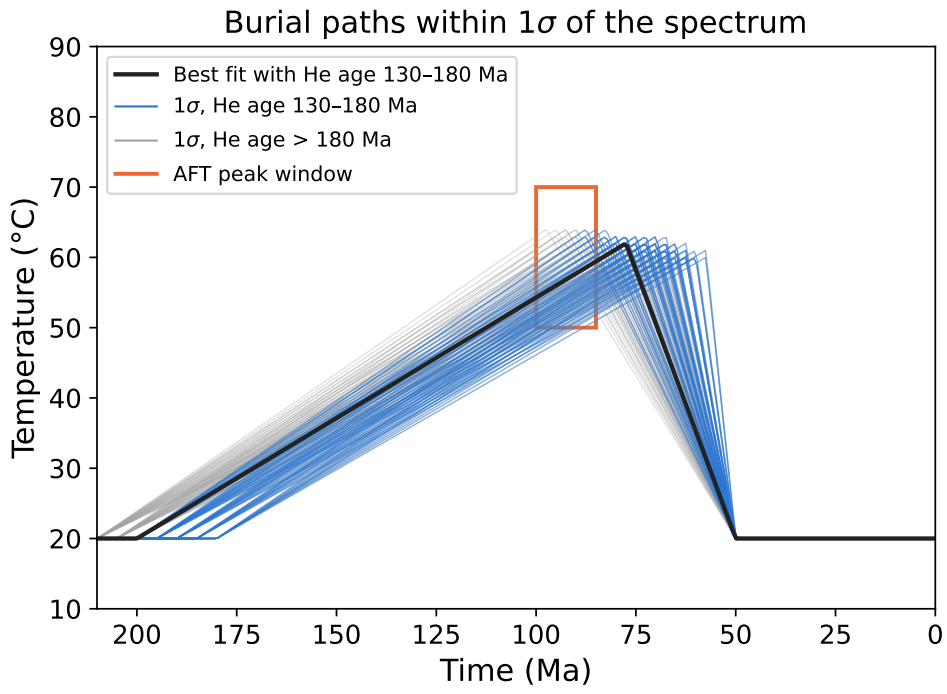


Figure 5: Thermal histories within the joint 1σ region of the spectrum fit. Blue paths also predict He ages within the measured 130–180 Ma range; gray paths predict older ages. The black path is the best-fitting age-consistent history, and the orange box is the apatite fission-track peak window.

and J. Zähringer, editors, *Potassium Argon Dating*, pages 68–107. Springer, Berlin, Heidelberg, 1966.

H. J. Gibson and K. Stüwe. Multiphase cooling and exhumation of the southern adelaide fold belt: constraints from apatite fission track data. *Basin Research*, 12(1):31–45, 2000. doi: 10.1046/j.1365-2117.2000.00110.x.

M. M. Mitchell, B. P. Kohn, and D. A. Foster. Post-orogenic cooling history of eastern south australia from apatite FT thermochronology. In Peter van den Haute and Frans de Corte, editors, *Advances in Fission-Track Geochronology: A selection of papers presented at the International Workshop on Fission-Track Dating, Ghent, Belgium, 1996*, pages 207–224. Springer Netherlands, Dordrecht, 1998. doi: 10.1007/978-94-015-9133-1_13.

D. L. Shuster and K. A. Farley. 4he/3he thermochronometry. *Earth and Planetary Science Letters*, 217:1–17, 2004. doi: 10.1016/S0012-821X(03)00595-8.

K. M. Waltenberg. Mineral physics and crystal chemistry of minerals suitable for weathering geochronology: Implications to 40ar/39ar and (u-th)/he geochronology, 2012. PhD thesis; institution not confirmed in the source draft.